



Akhuwat University

Department of Economics

B.S. 4 Years Program / Fifth Semester

Assignment 01: Advanced Mathematical Economics

Course Code: Econ 307

Note: Kindly ensure submission of the assignment by February 15, 2024, your adherence to this deadline is greatly appreciated, and feel free to contact me for any clarification or assistance.

Q. No. 1. Answer the following short questions:

(6x5=30)

- Linear vs Non-linear Differential Equations
- Homogeneous vs Non-Homogeneous
- Write and explain the general form of the first order linear differential equation.
- Keeping in view the general form of the first order linear differential equation, derive the general form of the first order linear differential equation with constant coefficient and constant term.
- Identify the various cases of the first order linear differential equation with constant coefficient and constant term.
- Differentiate between complementary function and particular integral.

Answer the following questions.

(3x10=30)

Q. No. 2. Consider the following first order linear differential equation with constant coefficient and constant term.

$$\frac{dy}{dt} + ay = b$$

- Find the general solution and definite solution if $a \neq 0$, $b=0$ (homogeneous case)
- Find the general solution and definite solution if $a \neq 0$, $b \neq 0$ (Non-homogeneous case)
- Find the general solution and definite solution if $a=0$, $b \neq 0$ (Non-homogeneous case)

Q. No. 3. Consider the following Differential equation:

$$\frac{dy}{dt} + 4y = 12; y(0) = 2$$

- Find complementary function y_c .
- Particular function y_p
- General Solution
- Definite Solution

Q. No. 4. Find the general and particular solution of the following differential equation by using an appropriate formula developed in Q. No. 2.

$$\frac{dy}{dt} - 5y = 0; y(0) = 6$$